

- Boolean Operations

An operation which combines 2 elements of the set to give another element of the same set is called a binary operations.

- Algebraic structure

A non-empty set equipped with one or more binary operations is called an algebraic structure.

eg: $(\mathbb{I}, +)$, $(\mathbb{R}, +, \cdot)$
set of integers.

- Group

An algebraic structure $(G, *)$ where (G) is a non empty set with the binary operation $(*)$, defined on it is said to be a group if the following properties are satisfied.

Properties are

G1: Closure Property:

$$a * b \in G \quad \forall a, b \in G$$

G2: Associative Property:

$$a * (b * c) = (a * b) * c \quad \forall a, b, c \in G$$

G3: Existence of Identity:

Let $e \in G$, then $a * e = a = e * a$, $\forall a \in G$
here, e is the identity element.

G4: Existence of Inverse:

If $a \in G$, $\exists b \in G$
such that $a * b = e$
then b is the inverse of a .

$\therefore (G, *)$ is a group as it holds all the properties

eg2: $(\mathbb{I}, +)$ is a group

$$G4: a + (-a) = 0$$

$-a$ is the inverse of a

here, it belongs to set of integers

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An algebraic structure that holds only closure property is called - Groupoid

eg: $(\mathbb{I}, -)$ is a groupoid as it is not associative but closure

* An algebraic structure that holds only closure & associative properties is called Semi group.
eg - $(\mathbb{N}, +)$

* An algebraic structure that holds ^{only} closure, associative & identity properties is called Monoid group.
eg - $(\mathbb{N}, +)$, $(\mathbb{Z}, +)$

Abelian Group

* Any group is called abelian group if it holds commutative property
 $\Rightarrow a * b = b * a \quad \forall a, b \in G$
eg) $(\mathbb{Z}, +)$ is an abelian group

Finite & Infinite Groups

* If we have finite no. of elements in the set: Finite group
eg - $G = (\{0, 1, 2, 3, 4\}, *)$

* Infinite group:

eg - $G = (\mathbb{Z}, +)$

Order of a Group

The no. of distinct elements in the set.
 $G = (\{0, 1, 2, 3, 4\}, *)$

$\Rightarrow$ Order of a group = 5

Ques - Let $\mathbb{Q}^+$ be the set of all positive rational numbers and $*$ is a binary operation defined on $\mathbb{Q}^+$ as $a * b = \frac{ab}{3}$.
Prove that $(\mathbb{Q}^+, *)$ is an abelian group.

Ans : Proof:

1) Closure Property: $a * b = \frac{ab}{3} \quad \forall a, b \in \mathbb{Q}^+$

$\frac{ab}{3} \in \mathbb{Q}^+ ; (\mathbb{Q}^+, *)$ is closed

2) Associative Property: $a * (b * c) = (a * b) * c$

L.H.S = $a * (b * c)$
 $= a * \left(\frac{bc}{3}\right)$

as:
 $a * b = \frac{ab}{3} \Rightarrow \frac{abc}{9}$

(LHS = RHS)

RHS = $(a * b) * c$
 $= \frac{ab}{3} * c$

$= \frac{abc}{9}$

* An algebraic structure that holds only closure & associative properties is called Semigroup.
eg- $(\mathbb{N}, +)$

* An algebraic structure that holds ^{only} closure, associative & identity properties is called Monoid group.
eg- $(\mathbb{N}, +)$, $(\mathbb{I}, \cdot)$

• Abelian Group-

If any group is called abelian group if it holds commutative property
 $\Rightarrow a * b = b * a \quad \forall a, b \in G$

eg) $(\mathbb{I}, +)$ is an abelian group

• Finite & Infinite Groups-

* If we have finite no. of elements in the set: Finite group
eg- $G = (\{0, 1, 2, 3, 4\}, *)$

* Infinite group:

$$G = (\mathbb{I}, +)$$

• Order of a Group-

no. of distinct elements in the set.
 $G = (\{0, 1, 2, 3, 4\}, *)$

Order of a group = 5

Ques- Let $\mathbb{Q}^+$ be the set of all positive rational no and $*$ is a binary operation defined on $\mathbb{Q}^+$ as $a * b = \frac{ab}{3}$.
Prove that $(\mathbb{Q}^+, *)$ is an abelian group.

Ans: Proof:

1) Closure Property: $a * b = \frac{ab}{3} \quad \forall a, b \in \mathbb{Q}^+$

$$\frac{ab}{3} \in \mathbb{Q}^+ ; (\mathbb{Q}^+, *) \text{ is closed}$$

2) Associative Property: $a * (b * c) = (a * b) * c$

$$\begin{aligned} \text{L.H.S} &= a * (b * c) \\ &= a * \left(\frac{bc}{3}\right) \end{aligned}$$

as:

$$a * b = \frac{ab}{3} \Rightarrow \frac{abc}{9}$$

$$\begin{aligned} \text{R.H.S} &= (a * b) * c \\ &= \frac{ab}{3} * c \end{aligned}$$

$$= \frac{abc}{9}$$

$$(\text{L.H.S} = \text{R.H.S})$$

3) Identity:

$$a * e = a$$

$$\frac{ae}{3} = a$$

$$ae - 3a = 0$$

$$a(e - 3) = 0 \quad a \neq 0$$

$$e - 3 = 0$$

$$\boxed{e = 3} \in \mathbb{Q}^+$$

3 is identity element.

5) Commutative property:

$$a * b = b * a$$

$$\frac{ab}{3} = \frac{ba}{3}$$

$$ab = ba$$

$(\mathbb{Q}^+, *)$ is an abelian group

* Composition Table for Finite Sets -

- A binary composition in a finite set can be shown in a tabular form known as composition table.
- The composition table for a finite group contains each element exactly once in each of its rows & columns.

Ques: Show that the set $\{1, -1, i, -i\}$ is an abelian group with respect to multiplication.

Proof: Composition Table:

$\times$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

1) Closure property

$\forall$ All the entries of the composition table are the element of the set. So it is a closed set or it holds closure property.

2) Associative property -

Take any 3 elements from the set.

$$a * (b * c) = (a * b) * c$$

$$\text{if } a = 1, b = -1, c = i$$

$$\text{LHS} = 1 * (-1 * i) = -i$$

$$\text{RHS} = (1 * -1) * i = -i$$

So, LHS = RHS, it holds associative property.

Identity property -

The headed row with one gives the same elements of the set that's why one is the identity element of the set. (Row $\rightarrow 1$) $\rightarrow$ see it.

1) Closure - it holds the closure property

2) Associative -

$$f_1 * (f_2 * f_3) = (f_1 * f_2) * f_3$$

$$\begin{aligned} \text{L.H.S} &= f_1 \circ (f_2 \circ f_3) \\ &= f_1 \circ f_4 \\ &= f_4 \end{aligned}$$

$$\begin{aligned} \text{R.H.S} &= (f_1 \circ f_2) \circ f_3 \\ &= f_2 \circ f_3 \\ &= f_4 \end{aligned}$$

$$\boxed{\text{L.H.S} = \text{R.H.S}}$$

3) Identity element -

f_1 is the identity element -

4) Inverse:

$$f_1^{-1} = f_1 \quad f_3^{-1} = f_3$$

$$f_2^{-1} = f_2 \quad f_4^{-1} = f_4$$

5) Commutative property -

Each row coincide with each column, so it holds commutative property.

Result -

$(G, *)$ is an abelian group

⇒ Order of an element:

Let G be a group & $a \in G$, then the order of an element a is the least positive integer n such that $\boxed{a^n = e}$. And it is called the order of this element.

If there exist no such n then the order of a is infinite or zero.

$$O(a) \leq O(G)$$

$$O(a^{-1}) = O(a)$$

Ques- $G = (\{1, \omega, \omega^2\}, \cdot)$ ω : unity
(cube root of unity) $\omega = \sqrt[3]{1} \Rightarrow \omega^3 = 1$

Ans, we know it is multiplication hence, 1 is the identity element.

$$O(1) : 1^1 = 1$$

$$O(1) = 1$$

$$O(\omega) : \omega^3 = 1$$

$$O(\omega) = 3$$

↳ order of ω

$$O(\omega^2) : (\omega^2)^3 = 1$$

$$O(\omega^2) = 3$$

$$\begin{cases} (\omega^2)^1 = \omega^2 \\ (\omega^2)^2 = \omega \\ (\omega^2)^3 = 1 \end{cases}$$

* Associative:

$$3 + (0 + 5) = (3 + 0) + 5$$

$$\text{L.H.S} = 3 + 5 = 8, \text{R.H.S} = 3 + 5 = 8$$

$$\text{L.H.S} = \text{R.H.S}$$

3) Identity:

(In addition, the identity element is zero).

0 is the identity element.

4) Inverse:

$$\begin{array}{ll} 0^{-1} = 0 & 3^{-1} = 3 \\ 1^{-1} = 5 & 4^{-1} = 2 \\ 2^{-1} = 4 & 5^{-1} = 1 \end{array}$$

5) Commutative:

Each row coincide with each column, so we can say that it is a commutative property here. [1st row & 1st column]

Thus $(G, +)$ is an abelian group

Q) Show that the finite set G has 4 elements -

$G = \{f_1, f_2, f_3, f_4\}$ where $f_1(x) = x$, $f_2(x) = -x$, $f_3(x) = \frac{1}{x}$, $f_4(x) = -\frac{1}{x}$, is an abelian group with $*$ is a binary operation that means composition of functions.

Proof - Composition Table

$*$	$f_1(x)$	$f_2(-x)$	$f_3(1/x)$	$f_4(-1/x)$
f_1	f_1	f_2	f_3	f_4
f_2	f_2	f_1	f_4	f_3
f_3	f_3	f_4	f_1	f_2
f_4	f_4	f_3	f_2	f_1

$$\begin{aligned} f_1 * f_1 &= f_1 \circ f_1 \\ &= f_1(f_1(x)) = f_1(x) \\ &= x = f_1 \end{aligned}$$

$$\begin{aligned} f_1 * f_2 &= f_1 \circ f_2 \quad (f_1 \circ f_2) \\ &= f_1(f_2(x)) = f_1(-x) \\ &= f_2 \end{aligned}$$

$$\begin{aligned} f_1 * f_3 &= f_1 \circ f_3 \\ &= f_1(f_3(x)) = f_1\left(\frac{1}{x}\right) \\ &= f_3 \end{aligned}$$

$$\begin{aligned} f_1 * f_4 &= f_1 \circ f_4 = f_1(f_4(x)) \\ &= f_1\left(-\frac{1}{x}\right) = f_4 \end{aligned}$$

$$f_2 * f_2 = f_2 \circ f_2$$

$$\begin{aligned} &= f_2(-x) \\ &= f_1(x) = f_1 \end{aligned}$$

4) Inverse Property:

$$(1)^{-1} = 1$$

$$(-1)^{-1} = -1$$

$$(c)^{-1} = -c$$

$$(-c)^{-1} = c$$

5) Commutative Property -

→ (Check all rows)

First row & First column (check)

So it is Abelian group

Addition Modulo m :

$a +_m b$ is written as:

where a & b are any integers and m is a fixed integer such that $a +_m b = r$, ($0 \leq r < m$) where, r is the least non-negative remainder where $\frac{a+b}{m}$ is divided by m .

eg- $8 +_{10} 3 = 1$

$$8 + 3 = 11$$

$$11 \div 10 = 1 \text{ (Remainder is 1)}$$

$6 +_8 2 = 0$

Multiplication Modulo m :

$$a \times_m b = r \quad (0 \leq r < m)$$

where a & b are any integers & m is a fixed integer such that $a \times_m b = r$, ($0 \leq r < m$) where, r is the least non-negative remainder when ab is divided by m .

eg- $8 \times_{12} 3 = 0$

$$\begin{aligned} 8 \times 3 &= 24 \\ 24 \div 12 &= 0 \end{aligned}$$

Show that $G = \{0, 1, 2, 3, 4, 5\}$ is a finite abelian group under addition modulo 6 (t_6)

Proof -

Composition Table - (C.T)

t_6	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

1) Closure Property of the C.T

• All these entries are the elements of the given set so, it holds a closure property.

Now, we are finding generators, (check by multiplication Modulo 9)

Check for 2 -
 $(2)^1 = 2$, $(2)^2 = 4$, $(2)^3 = 8$

$(2)^4 = 2 \times 2 \times 2 \times 2 = 7$ (solve: $24 \div 9 = 2 \text{ remainder } 6$)

$(2)^5 = 5$, $(2)^6 = 1$

$24 \div 9 = 2 \text{ remainder } 6$
 $16 \div 9 = 1 \text{ remainder } 7$
 $3 \div 9 = 0 \text{ remainder } 3$

2 is a generator of G

5 is also a generator of G as 5 is the inverse of 2 from previous page.

Check for 4
 $(4)^1 = 4$, $(4)^2 = 7$, $(4)^3 = 1$

4 is not a generator of G.

7 is also not a generator of G as 7 is the inverse of 4

Check for 8 -

$(8)^1 = 8$, $(8)^2 = 1$

8 is not a generator of G

Thus, 2 & 5 are the cyclic group only.

Ques- Show that the set $G = \{0, 1, 2, 3, 4\}$ is cyclic group. What are the generators.

Proof - C.T

+5	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

*) Closure -

All the entries of this C.T are in the set

*) Associative $(a * b) * c = a * (b * c)$; $a = 1, b = 2, c = 3$

*) Identity -

0 is the identity element & it is to set also

~~For~~ $-i$ is the inverse of i [Because it is inverse of i and it holds by the property of cyclic group.]

~~For~~ $(-i)$ is also a

$(-i)^1 = -i$, $(-i)^2 = -1$, $(-i)^3 = i$, $(-i)^4 = 1$
So, $(-i)$ is also a generator of G .

Notes:

If we are getting the identity element by the power less than the order of the group then it can't be a generator.

Q) Show that the group $G = \{1, 2, 4, 5, 7, 8\}$ & the binary operation is ^{multiplication modulo 9}

Q) Show that $G = (\{1, 2, 4, 5, 7, 8\}, \times_9)$ is a cyclic group. How many generators are there?

Soln:

$\times_9$	1	2	4	5	7	8
1	1	2	4	5	7	8
2	2	4	8	1	5	7
4	4	8	7	2	1	5
5	5	1	2	7	8	4
7	7	5	1	8	4	2
8	8	7	5	4	2	1

$$9 \nmid 14$$

Closure:

All the entries in the composition table are in

Set G

Associative: $(a * b) * c = a * (b * c)$, $a = 1, b = 2, c = 4$

$$(1 \times_9 2) \times_9 4 = 1 \times_9 (2 \times_9 4) = 1 \times_9 8 = 8$$

$$2 \times_9 4 = 1 \times_9 8 = 8$$

Inverse:

$$\begin{aligned} 1^{-1} &= 1, & 2^{-1} &= 5, & 3^{-1} &= 7, \\ 4^{-1} &= 2, & 7^{-1} &= 4, & 8^{-1} &= 8 \end{aligned}$$

this is an abelian group.

Identity element of the group can't be the generator.

$$Q) G = \{1, -1, i, -i\}$$

Ans - $O(1) = 1$

$$O(-1) = (-1)^2 = 1, O(-1) = 2$$

$$O(i) = (i)^4 = 1$$

$$O(i) = 4$$

$$O(-i) = (-i)^4 = 1$$

$$O(-i) = 4$$

→ Cyclic Group:

if there exists an element $a \in G$ for some group $(G, *)$ such that every element of G can be written as sum power of a i.e. a^n for some integer n then the group G is said to be cyclic group & the element a is called the generator of the group.

Properties:

1) Every cyclic group is an abelian group

2) If a is a generator of the cyclic group then a^{-1} is also a generator

3) If a is a generator of the cyclic group of order n then $a^n = e$ (identity element)

If G is a finite cyclic group of order n generated by a then a^m is also a generator of G if and only if m & n are relatively prime

Q) $G = \{1, -1, i, -i\}$ is a cyclic group w.r.t multiplication & also find the generator

$$O(1) = 1, (1)^2 = 1 \therefore 1 \text{ is not a generator of a group } G$$

$$O(-1) = -1, (-1)^2 = 1, (-1)^3 = -1 \therefore -1 \text{ is not a generator of group } G$$

$$O(i) = i, (i)^2 = -1, (i)^3 = -i, (i)^4 = 1$$

we get all the element of the set by the power of i . So, i is a generator of group G

Inverse

$$0^{-1} = 0$$

$$3^{-1} = 2$$

$$1^{-1} = 4$$

$$4^{-1} = 1$$

$$2^{-1} = 3$$

inverse exists

So, it is a group

Find min 1 generator to find

→

0 is not the generator as generator can't be the identity element

#

$$1^1 = 1$$

$$1^2 = 1 +_5 1 = 2$$

$$1^3 = 1 +_5 1 +_5 1 = 3$$

$$1^4 = 1 +_5 1 +_5 1 +_5 1 = 4$$

$$1^5 = 1 +_5 1 +_5 1 +_5 1 +_5 1 = 0$$

1 is a generator of group G .

as $1^{-1} = 4$

So, 4 is also a generator of G as 4 is the inverse of 1

$$2^1 = 2$$

$$2^2 = 2 +_5 2 = 4$$

$$2^3 = 2 +_5 2 +_5 2 = 1$$

$$2^4 = 2 +_5 2 +_5 2 +_5 2 = 3$$

$$2^5 = 2 +_5 2 +_5 2 +_5 2 +_5 2 = 0$$

2 is a generator of G

3 is also a generator of G as 3 is the inverse of 2.

Ques - let G is the group : $(G = \{1, \omega, \omega^2\}, \times)$ &
 $(G' = \{0, 1, 2\}, +_3)$ be two groups then show
 that $G \cong G'$

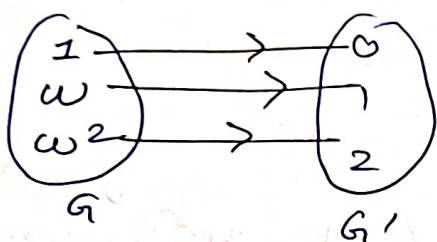
Ans - (If we have both finite set then
 make the composition Table)

$\times$	1	ω	ω^2
1	1	ω	ω^2
ω	ω	ω^2	1
ω^2	ω^2	1	ω

$+_3$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$$f(1) = 0, \quad f(\omega) = 1, \quad f(\omega^2) = 2$$

i) It is one one bec -



(f is one-one)

ii) onto -

As no element is left here so it is onto
 i.e. f is onto mapping.

Check - Homomorphism -

Consider any 2 elements from G .

$$f(1 \times \omega) = f(\omega)$$

$$= 1$$

$$= 0 +_3 1$$

$$= f(1) +_3 f(\omega)$$

This is the property of Homomorphism

So, it is isomorphism.

Isomorphism - Let $G \neq G'$ or

Let $(G, 0)$ & $(G', 0')$ be any two groups, then the mapping $f: G \rightarrow G'$ is called isomorphism if

- i) f is homomorphism
 - ii) f is one-one & onto.
- and G' is isomorphic image of G . $G \cong G'$ ↗ sign of isomorphism

Ques) Let R be the additive group of all Real numbers and R^+ be the multiplicative group of positive real numbers then the mapping - $f: R \rightarrow R^+$ is defined as ↳ homomorphism

$$f(x) = e^x \quad \forall x \in R$$

Show that $R \cong R^+$.

Proof - let $x, y \in R$

$$f(x) = e^x, \quad f(y) = e^y$$

$$f(x+y) = e^{x+y}$$

$$= e^x \cdot e^y$$

$$= f(x) \cdot f(y)$$

$\therefore f$ is homomorphism.

$$\text{let } f(x) = f(y)$$

$$e^x = e^y$$

Taking $\log$ both the sides

$$\boxed{x = y}$$

$\therefore$ this means f is one-one mapping

onto let $z \in R^+$, there exists an element z in its domain R so as in $\log(z)$ z should be the value

$$\log z \in R$$

$$f(\log z) = e^{\log z}$$

$$= z \in R^+$$

$\therefore f$ is onto mapping

Thus $R \cong R^+$, are isomorphic image of R

Ques-

If A is a generator of a group G of order 8. Then find the how many generators of this group & what are these.

Ans - $O(G) = 8$

Relatively prime numbers with 8 $2^3 = 8$

1, 3, 5, 7

[2 is not bec it is divisible by 8]

Generators: a^1, a^3, a^5, a^7

So, there are total 4 generators of the group G of order 8.

Homomorphism:

Let G and G' be any 2 groups with binary operations $\circ$ and $\circ'$ respectively. Then, the mapping $f: G \rightarrow G'$ is said to be Homomorphism if $\forall a, b \in G$ $f(a \circ b) = f(a) \circ' f(b)$

eg- Let $(G = \{1, -1, i, -i\}, \times)$ and I be the set of integers with addition. Now show that $f: I \rightarrow G$ is Homomorphism when $f(n) = i^n, \forall n \in I$

Proof: Let $m, n \in I$

$$f(m) = i^m, \quad f(n) = i^n$$

$$f(m+n) = i^{m+n}$$

$$= i^m \times i^n$$

$$f(m+n) = f(m) \times f(n)$$

② Hence, this mapping, f is called Homomorphism-

$$a+d = b+c \quad \text{and} \quad c+f = d+e$$

Adding

$$a+d+c+f = b+c+d+e$$

$$a+f = b+e$$

$$\Rightarrow (a, b) R (e, f)$$

$\therefore R$ is transitive

Hence, R is an equivalence Relation.

Q2) Show that the functions $f: \mathbb{R} \rightarrow (1, \infty)$ & $g: (1, \infty) \rightarrow \mathbb{R}$ defined as $f(x) = 5^{3x} + 1$ & $g(x) = \frac{1}{3} \log_5(x-1)$ are inverses

Proof: $f \circ g(x)$

$$\begin{aligned} \textcircled{1} f(g(x)) &= f\left(\frac{1}{3} \log_5(x-1)\right) \\ &= 5^{3 \left[\frac{1}{3} \log_5(x-1)\right]} + 1 \\ &= 5^{\log_5(x-1)} + 1 \\ &= x-1+1 \\ &= x \end{aligned}$$

$$\begin{aligned} \textcircled{2} g \circ f(x) &= g(f(x)) \\ &= g(5^{3x} + 1) \\ &= \frac{1}{3} \log_5(5^{3x} + 1 - 1) \\ &= \frac{1}{3} \log_5 5^{3x} \\ &= \frac{1}{3} \times 3x \log_5 5 \\ &= x \end{aligned}$$

means $f(x)$ is the inverse of $g(x)$

the inverse of $g(x)$

Q) Let $A = \{1, 2, 3, \dots, 5\}$ and R be the relation $A \times A$ defined by $(a, b) R (c, d)$ for $a+d = b+c$ in $A \times A$. Prove that R is an equivalence relation.

Ans: $A \times A = \{(1,1), (1,2), (1,3), (1,4), (1,5), (2,1), (2,2), (2,3), (2,4), (2,5), (3,1), (3,2), (3,3), (3,4), (3,5), (4,1), (4,2), (4,3), (4,4), (4,5), (5,1), (5,2), (5,3), (5,4), (5,5)\}$

$R = \{(1,3), (3,5), (4,6), (8,4), (2,4), (2,1), (5,1)\}$

Not equivalence.

Ans 1: Reflexive:

$(a, b) \in A \times A$
 $(a, b) R (a, b)$ if $a + b = b + a$
 $\therefore R$ is reflexive

Symmetric -

$(c, d) \in A \times A$
 $(a, b) R (c, d)$ then $(c, d) R (a, b)$
 (c, d) if $a + d = b + c$
 $\Rightarrow c + b = d + a$
 $\Rightarrow (c, d) R (a, b)$

R is symmetric

Transitive -

$(a, b), (c, d), (e, f) \in A \times A$
 $(a, b) R (c, d) \& (c, d) R (e, f)$

• Subgroup

A non empty subset H of a group G is called a subgroup of G if H itself is a group under the same binary operation.

2. Type of Subgroup

- Proper Subgroup
- Improper Subgroup

Every group G has atleast 2 subgroups which are G and e (Identity element), these are 2 subgroups are known as 'improper subgroups'.

• A sub-group other than these 2 is known as a proper subgroup

Probability

Counting Principle-

1. Sum Rule

2. Product Rule

→ if an event can be done in n_1 base & event 2 can be done in n_2 base & e_1 & e_2 are mutually exclusive then the no. of ways of both events occurring is $n_1 + n_2$ ways.

How many ways we can sum of 7 and 11 when 2 dice are rolled.

$$\text{Sum of 7 \& 11} = (n_1 + n_2)$$

$$(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1), (5, 6), (6, 5) = 11$$

Total no of ways $6 + 2 = 8$
Hence, mutually exclusive

AND → sum
OR → product

→ if two events are not mutually exclusive we apply product rules.

2.

Suppose a procedure with 2 disjoint subtasks if there are n_1 ways of doing the 1st task and (n_2) ways of doing the second then there are $n_1 \cdot n_2$ ways of doing the overall procedure.

d) A person has 4 hats, 3 pair of gloves & 5 pair of socks how many combinations can he try.

$$4 \cdot 3 \cdot 5 = 60$$

B) How many ways we can post 5 letters in 10 letter boxes.

→ 5 letters & 10 Boxes

$${}^{10}P_5 = \frac{10!}{5! \times 5!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5!}{5! \times 4! \times 3 \times 2 \times 1} =$$

• $UB \text{ of } A = \{18, 36\}$

$UB \text{ of } 9 = \{9, 18, 36\}$

$UB \text{ of } 18 = \{18, 36\}$

$UB \text{ of } \{9, 18\} = \{18, 36\}$

• $Lub \{9, 18\} = 18$

$\Rightarrow L.B \text{ of } 9 = \{9, 3, 1\}$

$L.B \text{ of } 18 = \{18, 9, 6, 3, 2, 1\}$

$LB \text{ of } A = \{9, 3, 1\}$

$\hookrightarrow$ (common find in 9 & 18)

$\therefore LB \text{ of } A = 9$

$\rightarrow$ same for another

8) Let $A = \{1, 2, 3\}$ and R be the relation on A defined as $R = \{(1, 1), (2, 2), (1, 2), (2, 3), (3, 3)\}$. Is R

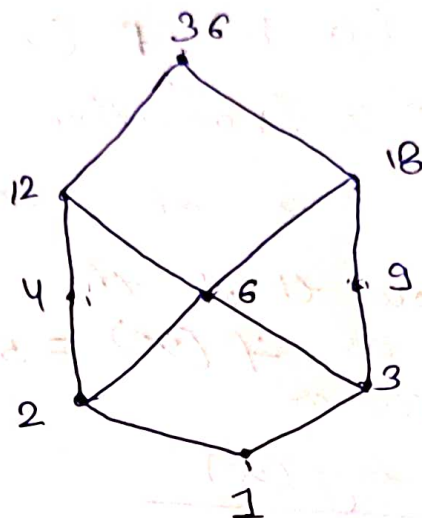
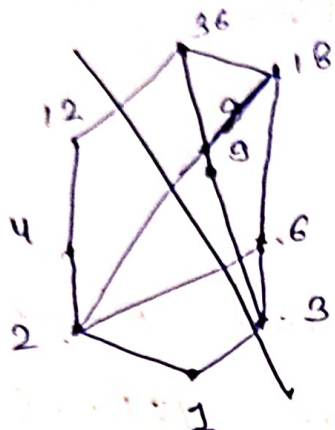
i) reflexive ii) symmetric iii) transitive
iv) Equivalence Relation.

Draw a diagram of R and also find the relation matrix M_R . And R^2 .

Q3) Draw the Hasse Diagram of D_{36}

$$D_{36} = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$$

Check prime no.



Q) $A = \{9, 18\}$

$B = \{4, 6, 9\}$

Find LUB & GLB of A & B .

Find LUB & GLB of A & B .

$\text{LUB} = \{36, 18\}$

$\text{GLB} = \{9\}$

$\text{LUB} = \{18, 3\}$

$\text{GLB} = \{9\}$

$\{2, 4, 6, 9\} = \{36\}$

$\{2, 4, 6, 9\} = \{18\}$

$\text{LUB} = \{36\}$

$\text{GLB} = \{1\}$

1 letter can be posted in 10 ways
Hence 5 letters can be posted
 $= 10^5$

1 letter can be posted in 10 ways
Hence 5 letters can be posted
 $= 10^5$

- ~~None~~ digits are repeated.
- digits can be repeated.

ii) $3 \times 3 \times 3 \times 3 \times 3 = 3^5 + 2 \times 2 \times 2 \times 2 \times 2 = 1$

$$\begin{aligned} &= 5 \times 4 \times 3 + 5 \times 4 + 1 \times 5 \\ &= 60 + 20 + 5 \\ &= 85 \end{aligned}$$

$$\begin{aligned} 1) \quad & 5 \times 5 \times 5 \times 5 \times 5 + 5 \times 5 + 5 \\ &= 5 + 125 + 25 \\ &= 155 \end{aligned}$$

1) A store has 8 regular doors

Permutations -

1st yr 10 (Group 2)

The no. of permutation of n -different object taken r at a time is represented as ${}^n P_r$

$${}^n P_r = \frac{n!}{(n-r)!}$$

eg - Total objects
 n, y, z

$${}^3 P_2 = \frac{3!}{(3-2)!} = 3 \times 2 = 6$$

xy, yx, yz, zy, xz, zx

$${}^3 P_3 (3 \text{ at a time}) = \frac{3!}{0!} = 3!$$

If we have n object & we have to take n object at a time: ${}^n P_n = n!$ (all things together)

3) In how many ways can 5 children stand in a queue - $\rightarrow$ take the arrangement
 ${}^5 P_5 = 5! = 120$

4) In how many ways can 5 English, 4 Tamil & 2 Hindi books can be arranged, if the books of each different language are kept together

$$\begin{aligned} & {}^5 P_5 \times {}^4 P_4 \times {}^2 P_2 \\ &= 5! \times 4! \times 2! \times 3! \\ &= 120 \times 24 \times 2 \times 3 \times 1 \\ &= 1728 \end{aligned}$$

6 arrangement as we can shuffle it ${}^3 P_3$ times 2^4

120
24

Q) How many 4-digit nos are there with distinct digits

Ans) ${}^{10}P_4 = \frac{10!}{(10-4)!} = \frac{10!}{6!}$

$= \frac{10 \times 9 \times 8 \times 7 \times 6!}{6!}$

(i) ${}^{10}P_4 = 5040$ ways

we can't place 0 here as it'll become 3 digit no

${}^9P_4 = 9 \times 8 \times 7 \times 6 = 9 \times 8 \times 7 \times 6$
 (1 entry will be from 9)

(ii)

or another ways:
 Total 10 digit

we can't place zero here that's why 9 digit
 ${}^9P_1 \times {}^8P_1 \times {}^7P_1 \times {}^6P_1 = 9 \times 8 \times 7 \times 6$

Permutation of things not all different.

The no. of permutations of n things taken r at a time in which k_1 elements are of one kind, k_2 elements are of second kind, ...

$${}^n P_r = \frac{n!}{k_1! k_2! k_3! \dots k_n!}$$

Q1) How many different arrangement can be made by using 4 red and 3 black identical balls.

~~${}^7P_4 \times {}^7P_3$~~

Ans - Total no. of balls = 7

Total arrangements = $\frac{11!}{4! \times 3!}$

Q2) In how many of the different permutations of the letters of the word "MATHEMATICS", vowels occurred together.

Ans Total vowels in this word MATHEMATICS is -
A A E I 2 repeated (2!)

Total consonants to lie r = 8 (each word unit)

$$= \frac{8!}{2! \times 2!} \quad \text{--- (i)}$$

For the arrangement of vowels $\frac{4!}{2!} \rightarrow$ (A repeats 2 times)

Total arrangement = eq (i) $\times$ eq (ii)

$$= \left(\frac{8!}{2! \times 2!} \right) \times \left(\frac{4!}{2!} \right)$$

$$= \frac{8!}{2!}$$

Q3) In how many different ways, we can arrange the letters of the word "TABLET" so that the middle position is always occupied by T.

$$\frac{4!}{4!} \frac{T}{1} \frac{4!}{4!} = 4P_4 = 4!$$

$$4 \times 1 \times 1 \times 4 \times 4$$

Q) In how many different ways we can arrange the letter of the word CAPTOP so that it will start & end with vowels.

Ans -

$$\text{Total ways} = 2 \times 4! = 48 \text{ ways}$$

$$\begin{matrix} \overline{C} & \overline{A} & \overline{T} & \overline{O} & \overline{P} \\ (2!) & & & & (2!) \end{matrix}$$

$$\frac{4!}{2! \times 2!}$$

Q) In how many diff. ways can 4 boys & 5 girls be seated in a row so that they are alternate

Ans -

$$\underline{G_1} \quad \underline{B_1} \quad \underline{G_2} \quad \underline{B_2} \quad \underline{G_3} \quad \underline{B_3} \quad \underline{G_4} \quad \underline{B_4} \quad \underline{G_5}$$

$$\begin{aligned} \text{Total no. of arrangements} &= 5! \times 4! \\ &= 5 \times 4 \times 3 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \\ &= 2880 \text{ ways} \end{aligned}$$

$$\begin{array}{r} 24 \\ \times 48 \\ \hline 24 \times 8 \\ 24 \times 8 \\ \hline 288 \end{array}$$

• If there are 5 Boys & 5 Girls (alternate pos)

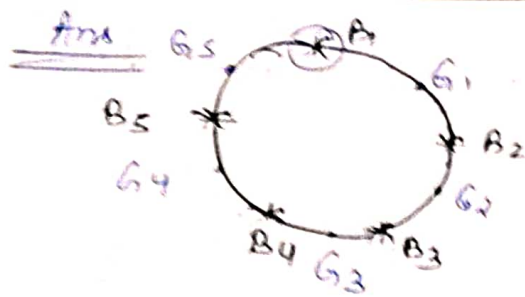
$$\begin{matrix} \underline{G_1} & \underline{B_1} & \underline{G_2} & \underline{B_2} & \underline{G_3} & \underline{B_3} & \underline{G_4} & \underline{B_4} & \underline{G_5} & \underline{B_5} \\ B_1 & G_1 & B_2 & G_2 & B_3 & G_3 & B_4 & G_4 & B_5 & G_5 \end{matrix} \rightarrow 2 \text{ arrangement are possible}$$

$$\text{Total ways} = \underset{\substack{\downarrow \\ \text{(for boys)}}}{5!} \times \underset{\substack{\downarrow \\ \text{(for girls)}}}{5!} \times 2 \rightarrow (2 \text{ ways for sitting})$$

Circular Permutation -

For n persons, (circular table)
Total arrangement
= $(n-1)!$

Q) In how many diff. ways can 5 boys & 5 girls be seated around the circular table so that the boys & girls are at alternate position.



- ① $B_1 \rightarrow$ fixed
4 boys $\rightarrow$ 4 position $\rightarrow 4!$
girls $\rightarrow 5 \rightarrow$ 5 position $\rightarrow 5!$

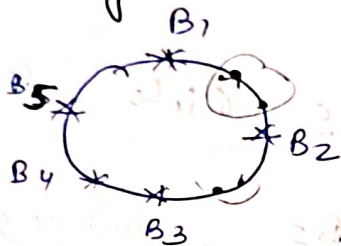
- ② $G_1 \rightarrow$ fixed
(vice-versa)

Total no. of arrangement = $4! \times 5!$

- Q) 4 boys & 5 girls (same as prev. ques.)
It is not possible in circular table.

- Q) There are 5 boys & 4 girls to sit in a circular table. In how many ways can they sit so that no two girls will sit together. Find total no. of arrangement.

Ans-



Total no. of ways = ~~5P4~~ $\rightarrow$ 5 places to sit for 4 girls
 $5 \cdot 4! \times 5P_4!$
boys girls

Combinations:

If we have to select: $nC_r = \frac{n!}{r!(n-r)!}$

① $nC_n = 1$, ② $nC_0 = 1$, ③ $nCr = nC_{n-r}$

- Q) How many different selection of 6 books can be made from 11 different books if 2 particular books are always selected.

Total book = 11

Remaining books from selected = 9

Total selected = 4

$$r^2 - 3r + 2 = 0 \quad // \text{Auxiliary Equation}$$

Case 1: When the roots are distinct - $r_1, r_2, r_3, \dots$

$$a_n = C_1 r_1^n + C_2 r_2^n + C_3 r_3^n + \dots$$

Case 2: When the roots are repeated: $r_1 = r_2, r_3, r_4, \dots$

$$a_n = (C_1 + n C_2) r^n + C_3 r_3^n + \dots$$

$$a_n = (C_1 + n C_2 + n^2 C_3) r^n + C_4 r_4^n + \dots$$

Case 3: When the roots are imaginary $a \pm ib$

$$a_n = C_1 (a+ib)^n + C_2 (a-ib)^n + \dots$$

Solve: A.E: $r^2 - 3r + 2 = 0$
 $(r-2)(r-1) = 0$
 $r = 2, 1$

So, recurrence relation -

$$a_n = C_1 + C_2 \cdot 2^n$$

Ques: $a_n = 3a_{n-1} - 4a_{n-2}$,
 $a_0 = 1, a_1 = 2$ (To find the value of the constant)

Ans: $a_n - 3a_{n-1} - 4a_{n-2} = 0$

Order = 2

Put $a_n = r^n$

$$r^n - 3r^{n-1} - 4r^{n-2} = 0$$

$\div$ by

$$r^{n-2}$$

$$r^2 - 3r - 4 = 0$$

$$r^2 - 4r + r - 4 = 0$$

Recurrence Relations

A formula which defines any term of a sequence in terms of any no. of its prev. previous terms is called recursive formula & the relation is called the Recurrence Relation.

eg

3, 8, 13, 18, - - - - -

$$a_n = a_{n-1} + 5, \quad n \geq 2$$

$$a_1 = 3$$

Recurrence
relation

$$F(a_n, a_{n-1}, a_{n-2}, \dots, a_{n-k})$$

(order = k)

eg - $a_n - 3a_{n-1} + 2a_{n-2} = 0$

(least term)

$$\text{Order} = 2$$

Methods

① Iteration Method - (Not used)

② Characteristic Roots Method

③ Generating function Method

1) Characteristic Roots Method -

eg - $a_n - 3a_{n-1} + 2a_{n-2} = 0$

$$\text{order} = 2$$

$$\text{put } a_n = r^n$$

$$r^n - 3r^{n-1} + 2r^{n-2} = 0$$

divided by r^{n-2} ;

9C_4 - (9 books mai se 4 books select krni hai)

Q) There are 4 black, 5 red & 6 green balls. We have to select 3 balls in how many ways we can select

i) All balls are of different colours. ${}^{19}C$

(+) → task complete

(*) → task not fully completed

$${}^4C_1 * {}^5C_1 * {}^6C_1$$

(ii) All balls are of same colour -

$${}^4C_3 + {}^6C_3 + {}^5C_3$$

(iii) No red ball -

$${}^{10}C_3$$

(iv) At least one is a red ball -

$${}^5C_1 \times {}^{10}C_2 + {}^5C_2 \times {}^{10}C_1 + {}^5C_3$$

or

Total ways - (to select the no red balls)
= at least one red ball.

$$\text{Total ways} - \text{No red balls} = \text{at least one red ball}$$

$${}^{15}C_3 - {}^{10}C_3 =$$

(v) Exactly one is green ball.

$$= {}^6C_1 \times {}^9C_2$$

Homogeneous soln -

$$y_k^{(h)} = C_1 3^k + C_2 (-2)^k$$

$$\text{Let } y_k^{(h)} = A$$

From eqn (1) -

$$A - A - 6A = -30$$

$$-6A = -30$$

$$\boxed{A = 5}$$

$$y_k^{(p)} = 5$$

$$\text{Soln - } y_k = y_k^{(h)} + y_k^{(p)}$$

$$\boxed{y_k = C_1 3^k + C_2 (-2)^k + 5}$$

$$y_0 = C_1 3^0 + C_2 (-2)^0 + 5$$

$$20 = C_1 + C_2 + 5$$

$$C_1 + C_2 = 15 \quad \text{--- (2)}$$

$$y_1 = 3C_1 - 2C_2 + 5$$

$$-5 = 3C_1 - 2C_2 + 5$$

$$3C_1 - 2C_2 = -10 \quad \text{--- (3)}$$

By solving (2) & (3) -

$$C_1 = 4, \quad C_2 = 11$$

$$\underline{\text{Soln}} - y_k = 4 \cdot 3^k + 11(-2)^k + 5 \quad \text{--- (Ans)}$$

Solution of Non - Homogeneous Recurrence Relation -

$$a_n + c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} = f(n)$$

Solution - $a_n = a_n^{(h)} + a_n^{(p)}$

h = homogenous solⁿ

p = particular solⁿ

Trial Funcⁿ

$A \quad Ab^n \quad A_0 n^s c^n \quad A_0 + A_1 n + A_2 n^2 + \dots + A_m n^m$
 $(A_0 + A_1 n + A_2 n^2 + \dots) b^n$

$f(n)$

- ① Constant
- ② b^n (b is not the characteristic root).
- ③ c^n (c is the characteristic root with multiplication)
- ④ $p(n)$ Poly of degree m .
- ⑤ $b^n p(n)$ (b is not the characteristic root)

Ques - $y_k - y_{k-1} - 6y_{k-2} = -30$ — ①,

$y_0 = 20, \quad y_1 = 5$

$\Rightarrow y_k - y_{k-1} - 6y_{k-2} = 0$

Put $y_k = r^k$

$r^k - r^{k-1} - 6r^{k-2} = 0$

divide by r^{k-2}

A. E : $r^2 - r - 6 = 0$
 $(r - 3)(r + 2) = 0$
 $(r = 3, -2)$

$$r(r-4)+1(r-4)=0$$

$$(r+1)(r-4)=0$$

$$(r=-1, r=4)$$

$$A.E: r^2 - 3r - 4 = 0$$

$$r = -1, 4$$

(By recurrence relation)

$$a_n = (-1)^n C_1 + C_2 4^n$$

$$a_0 = (-1)^0 C_1 + C_2 (4)^0$$

$$1 = C_1 + C_2$$

$$\boxed{C_1 = 1 - C_2} \quad (1)$$

$$a_1 = (-1)^1 C_1 + C_2 (4)$$

$$2 = -C_1 + 4C_2$$

$$4C_2 = 2 + C_1$$

$$\left(\frac{C_2 = C_1 + 2}{4} \right) \Rightarrow$$

$$4C_2 = \frac{1 - C_2 + 2}{1}$$

$$4(C_2 + C_2) = 1$$

$$5C_2 = 1$$

$$C_2 = \frac{1}{5}$$

$$4C_2 = 3 - C_2$$

$$5C_2 = 3$$

$$\left(C_2 = \frac{3}{5} \right) \quad (2)$$

Put in eq (1) the value of (2)

$$C_1 = 1 - \frac{3}{5} = \frac{2}{5}$$

$$\boxed{C_1 = 2/5}$$

From eqⁿ ①

$$A_0 + A_1(x+2) + A_2(x+2)^2 = 5[A_0 + A_1(x+1) + A_2(x+1)^2] +$$

$$A_0 + A_1(x+2) + A_2(x+2)^2 = 5[A_0 + A_2(x+1) + A_2(x+1)^2] + 6(A_0 + A_1x + A_2x^2) = x^2 + 2x + 1$$

$$A_0 + xA_1 + 2A_1 + A_2(x^2 + 4 + 2x) = 5A_0 + (-5A_1x) - 5 - 5A_2(x^2 + 1 + 2x) + 6A_0 + 6A_1x + 6A_2x^2 = x^2 + 2x + 1$$

$$01 = A$$

$$04 = 10x + 40$$

$$0x = 01 + 0 + 0 + 0 = 0$$

initially we have to compare relation
 $\Delta x = 1 - x = 0$
 ①

$$C_1 + C_2 = 12 \quad \text{--- (2)}$$

$$a_1 = C_1 + C_2 - 27 = 7$$

$$2C_1 + 4C_2 = 34 \quad \text{--- (3)}$$

$$\begin{array}{r} 2C_1 + 2C_2 = 24 \\ 2C_1 + 4C_2 = 34 \\ \hline -2C_2 = -10 \end{array}$$

$$\boxed{C_2 = 5}$$

$$C_2 + 5 = 12$$

$$\boxed{C_2 = 7}$$

The soln is-

$$a_n = 7 \cdot 2^n + 5 \cdot 4^n - 9 \cdot 3^n$$

Ques - $a_{r+2} - 5a_{r+1} + 6a_r = (r+1)^2 \quad \text{--- (1)}$

$$a_{r+2} - 5a_{r+1} + 6a_r = 0$$

Put $a_r = m^r$ ---

$$m^{r+2} - 5m^{r+1} + 6m^r = 0$$

divided by m^r ---

$$A-E = m^2 - 5m + 6 = 0$$

$$\text{roots} = 2, 3$$

$$a_r(h) = C_1 2^r + C_2 3^r$$

Let, $a_r(g) = A_0 + A_1 r + A_2 r^2$

Ques- $a_n - 6a_{n-1} + 8a_{n-2} = 3^n$ — (1),
 $a_0 = 3, a_1 = 1$

$$\Rightarrow a_n - 6a_{n-1} + 8a_{n-2} = 0$$

Put $a_n = r^k$ —

$$r^k - 6r^{k-1} + 8r^{k-2} = 0$$

divide by r^{k-2} —

A.E: $r^2 - 6r + 8 = 0$

$$r^2 - 4r - 2r + 8 = 0$$

$$r(r-4) - 2(r-4) = 0$$

$$(r-4)(r-2) = 0$$

$$\boxed{r = 2, 4}$$

$$\boxed{a_n^{(h)} = C_1 2^n + C_2 4^n}$$

Let, $a_n^{(p)} = A b^n = A 3^n$

Putting in eqⁿ (1) $\rightarrow$

$$A 3^n - 6A 3^{n-1} + 8A 3^{n-2} = 3^n$$

$$A \cdot 3^{n-2} (9 - 18 + 8) = 3^n$$

$$-A 3^{n-2} = 3^{n-2} \times 9$$

$$\boxed{A = -9}$$

$$a_n^{(p)} = -9 \cdot 3^n$$

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$a_n = C_1 2^n + C_2 4^n - 9 \times 3^n$$

$$a_0 = C_1 + C_2 - 9 = 3$$

$$Q) a_r - 9a_{r-1} + 20a_{r-2} = 25^r \text{ --- (1)}$$

$$a_r - 9a_{r-1} + 20a_{r-2} = 0$$

$$\text{Put } a_r = m^r$$

$$m^r - 9m^{r-1} + 20m^{r-2} = 0$$

$$\text{Divided by } m^{r-2}$$

$$\text{A.E. - } m^2 - 9m + 20 = 0$$

$$\text{roots} = 4, 5$$

$$a_r^{(h)} = C_1 4^r + C_2 5^r$$

$$\text{let } a_r^{(p)} = A_0 r 5^r$$

From eq (1)

$$A_0 r 5^r - 9A_0 (r-1) 5^{r-1} + 20A_0 (r-2) 5^{r-2} = 25^r$$

Find the value of A_0 -

$$25 A_0 r 5^{r-2} - 9 \times 5 A_0 (r-1) 5^{r-2} + 20 A_0 (r-2) 5^{r-2} = 25^{r-2} \times 25$$

$$25 A_0 r - 45 A_0 r + 45 A_0 + 20 A_0 r - 40 A_0 = 50$$

$$5 A_0 = 50$$

$$\boxed{A_0 = 10}$$

$$\boxed{a_r^{(p)} = 10r 5^r}$$

$$\text{Soln is: } a_r = a_r^{(h)} + a_r^{(p)}$$

$$\boxed{a_r = C_1 4^r + C_2 5^r + 10r 5^r} \text{ --- Ans}$$

Ques) Solve this recurrence relation -

$$a_r - 7a_{r-1} + 12a_{r-2} = r \cdot 4^r \text{ --- (1)}$$

$$ar - 7ar-1 + 12ar-2 = 0$$

$$\boxed{\text{Put } ar = m^r -}$$

$$m^r - 7m^{r-1} + 12m^{r-2} = 0$$

$$\boxed{\text{divided by } m^{r-2}}$$

$$m^2 - 7m + 12 = 0$$

$$m^2 - 4m - 3m + 12 = 0$$

$$m(m-4) - 3(m-4) = 0$$

$$(m-4)(m-3) = 0$$

$$m = 4, 3$$

$$a_r^{(h)} = c_1 3^r + c_2 4^r$$

$$\text{let ; } a_r^{(p)} = (A_0 + A_1 r) \cdot 4^r$$

From eqⁿ ① -

$$(A_0 + A_1 r) \cdot 4^r - 7(A_0 + A_1 r)(r-1) 4^{r-1} + 12(A_0 + A_1 r)(r-2) 4^{r-2} = r \cdot 4^r$$

$$(A_0 + A_1 r) \cdot 4^r - 7[(A_0 + A_1(r-1))(r-1) 4^{r-1}] + 12[(A_0 + A_1(r-2))(r-2) 4^{r-2}] = r \cdot 4^r$$

$$4^2 \cdot (A_0 + A_1 r) \cdot 4^{r-2} - 7 \times 4 [A_0 + A_1(r-1)](r-1) 4^{r-2} + 12 [A_0 + A_1(r-2)](r-2) 4^{r-2} = r \cdot 4^{r-2} \cdot 4^2$$

Generating Function-

$$G(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

$$= \sum_{k=0}^{\infty} a_k x^k$$

a_k	$G(x)$
1	$\frac{1}{1-x}$
$k+1$	$\frac{1}{(1-x)^2}$
k	$\frac{x}{(1-x)^2}$
a^k	$\frac{1}{1-a^x}$
$k(k+1)$	$\frac{2x}{(1-x)^3}$
$(k+1)(k+2)$	$\frac{2}{(1-x)^3}$

Ques) $a_{r+2} - 3a_{r+1} + 2a_r = 0$, $a_0 = 2, a_1 = 3$
 Solve this recurrence relation by generating function
 $\Rightarrow$ it is homogeneous funcⁿ

Solⁿ: Multiply by x^r & do sum 0 to ∞

$$\sum_{r=0}^{\infty} a_{r+2} x^r - 3 \sum_{r=0}^{\infty} a_{r+1} x^r + 2 \sum_{r=0}^{\infty} a_r x^r = 0$$

$$a_2 + a_3 x + a_4 x^2 + \dots - 3(a_1 + a_2 x + a_3 x^2 + \dots) + 2(a_0 + a_1 x + a_2 x^2 + \dots) = 0$$

$\times 2 \div \text{by } x$

$$\frac{1}{x^2} (a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots) - \frac{3}{x} (a_1 x + a_2 x^2 + a_3 x^3 + \dots) + 2 G(x) = 0$$

From generating function : eqn

$$\frac{1}{x^2} (G(x) - a_0 - a_1 x) - \frac{3}{x} [G(x) - a_0] + 2 G(x) = 0$$

$$\frac{G(x) - 2 - 3x}{x^2} - \frac{3[G(x) - 2]}{x} + 2 G(x) = 0$$

$$G(x) - 2 - 3x - 3x G(x) + 6x + 2x^2 G(x) = 0$$

$$(2x^2 - 3x + 1) G(x) + 3x - 2 = 0$$

$$G(x) = \frac{2 - 3x}{2x^2 - 3x + 1}$$

$$G(x) = \frac{2 - 3x}{(2x-1)(x-1)}$$

By partial fraction -

$$\frac{A(2-3x)}{(2x-1)(x-1)} = \frac{A}{(2x-1)} + \frac{B}{(x-1)} \quad (1)$$

$$2 - 3x = A(x-1) + B(2x-1)$$

if $x=1$

$$2 - 3 = 0 + B(2 - 1)$$

$$\boxed{B = -1}$$

$$\text{if } x = \frac{1}{2} \quad \left| \begin{array}{l} 2 - \frac{3}{2} = A\left(-\frac{1}{2}\right) \\ \frac{1}{2} = A - \frac{1}{2} \end{array} \right.$$

$$\frac{1}{2} = A - \frac{1}{2}$$

$$\boxed{A = -1}$$

Put in eqn ①

$$G(x) = \frac{-1}{2x-1} = \frac{1}{1-2x}$$

Convert into (1-x) form

$$G(x) = \frac{1}{1-2x} = \frac{1}{1-x}$$

$$\sum a_r x^r = \sum 2^r \cdot x^r + \sum 1 \cdot x^r$$

$$\boxed{a_r = 2^r + 1} \text{ Ans}$$

Ques - $a_r - 7a_{r-1} + 10a_{r-2} = 0$ & $a_0 = a_1 = 3$

Solve this recurrence relation by generating function

Multiply by x^r & do sum 2 to ∞

$$\sum_{r=2}^{\infty} a_r \cdot x^r - 7 \sum_{r=2}^{\infty} a_{r-1} \cdot x^r + 10 \sum_{r=2}^{\infty} a_{r-2} \cdot x^r = 0$$

$$(a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots) - 7(a_1 x^2 + a_2 x^3 + a_3 x^4 + \dots) + 10(a_0 x^2 + a_1 x^3 + a_2 x^4 + \dots) = 0$$

$$\begin{aligned} \text{by } G(x) &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots \\ &= \sum_{k=0}^{\infty} a_k x^k \end{aligned}$$

$$G(x) - a_0 - a_1 x - 7x[a_1 + a_2 x + a_3 x^2 + \dots] + 10x^2[a_0 + a_1 x + a_2 x^2 + \dots] = 0$$

$$G(x) - 3 - 3x - 7x[G(x) - 3] + 10x^2 G(x) = 0$$

$$G(x)(10x^2 - 7x + 1) - 18x + 3 = 0$$

$$G(x) = \frac{3 - 18x}{10x^2 - 7x + 1}$$

$$10n^2 - 7n + 1$$

$$\quad \quad \quad \begin{matrix} -5n & -2n \end{matrix}$$

$$10n^2 - 5n - 2n + 1$$

$$\text{Factor}$$

$$10n^2 - 2n - 5n + 1$$

$$2n(5n-1) - 1(5n-1)$$

$$(2n-1)(5n-1)$$

$$G(n) = \frac{3-18n}{(5n-1)(2n-1)}$$

$$\frac{3-18n}{(5n-1)(2n-1)} = \frac{A}{5n-1} + \frac{B}{2n-1}$$

$$3-18n = A(2n-1) + B(5n-1)$$

$$\text{if } 2n-1=0$$

$$n=1/2$$

$$3-9 = B\left(\frac{5}{2}-1\right)$$

$$-6 = B\left(\frac{5-2}{2}\right)$$

$$-\frac{6}{1} = B\left(\frac{3}{2}\right)$$

$$(B = -4)$$

$$\text{if } 5n-1=0$$

$$n=1/5$$

$$3-\frac{18}{5} = A\left(\frac{2}{5}-1\right)$$

$$\frac{15-18}{5} = A\left(\frac{2-5}{5}\right)$$

$$-\frac{3}{5} = A\left(-\frac{3}{5}\right)$$

$$(A = 1)$$

$$G(n) = \frac{1}{5n-1} - \frac{4}{2n-1}$$

$$G(n) = \frac{-1}{1-5n} + \frac{4}{1-2n}$$

$$a_r = -5^r + 4 \cdot 2^r$$

Ques) The recurrence relation is -
 $a_{r+2} - 2a_{r+1} + a_r = 2^r$, & initial conditions
 $a_0 = 2, a_1 = 1$

Sol) Multiply by x^r & sum @ to ∞

$$\sum_{r=0}^{\infty} a_{r+2} \cdot x^r - 2 \sum_{r=0}^{\infty} a_{r+1} \cdot x^r + \sum_{r=0}^{\infty} a_r \cdot x^r = \sum_{r=0}^{\infty} 2^r \cdot x^r$$

$$(a_2 + a_3x + a_4x^2 + \dots) - 2(a_1x + a_2x^2 + a_3x^3 + \dots) + (a_0 + a_1x + a_2x^2 + \dots) = \frac{1}{1-2x}$$

(x & ÷ by x^2)

$$\frac{1}{x^2} [a_2x^2 + a_3x^3 + a_4x^4 + \dots] - \frac{2}{x} [a_1x + a_2x^2 + a_3x^3 + \dots] + (a_0 + a_1x + a_2x^2 + \dots) = \frac{1}{1-2x}$$

$$\frac{G(x) - a_0 - 2 - x}{x^2} - \frac{2(G(x) - 2)}{x} + G(x) = \frac{1}{1-2x}$$

$$G(x) - 2 - x - 2xG(x) + 4x + x^2G(x) = \frac{x^2}{1-2x}$$

$$(x^2 - 2x + 1)G(x) + 3x - 2 = \frac{x^2}{1-2x}$$

$$G(x) [x^2 - 2x + 1] = \frac{x^2}{1-2x} - 3x + 2$$

$$G(x) = \frac{x^2}{(1-2x)(x^2-2x+1)} - \frac{3x+2}{(x^2-2x+1)} = \frac{x^2 - (1-2x)(3x+2)}{(x^2-2x+1)(1-2x)}$$

$$G(x) = \frac{x^2 - [3x^2 + 2 - 6x^2 + 4x]}{(1-2x)(x-1)^2} = \frac{x^2 - [3x^2 - 6x^2 + 4x - 2]}{(1-2x)(x-1)^2} = \frac{x^2 - [3x^2 - 6x^2 + 4x - 2]}{(1-2x)(x-1)^2}$$

$$G(x) = \frac{7x^2 + 7x - 2}{(1-2x)(x-1)^2}$$

by partial fraction -

$$\begin{aligned} 1-2x &= 0 \\ 2x &= 1 \\ x &= 1/2 \end{aligned}$$

$$\frac{7x^2 + 7x - 2}{(1-2x)(x-1)^2} = \frac{A}{1-2x} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$$

$$7x^2 + 7x - 2 = A(x-1)^2 + B(1-2x)(x-1) + C(1-2x)$$

if $x = 1$

$$7 - 7 - 2 = B(1-2)(1-1) + C(1-2)(1-1)$$

$$\boxed{C = -2}$$

if $x = 1/2$

$$7\left(\frac{1}{4}\right) + \frac{7}{2} - 2 = A\left(-\frac{1}{2}\right)\left(\frac{1}{4}\right)$$

$$\frac{7 + 14 - 8}{4} = A\left(\frac{-1}{8}\right)$$

$$13 = A(-2)$$

$$1 = A\left(\frac{-1}{2}\right)$$

$$\boxed{A = -2}$$

$$7x^2 - 7x + 2 = -2(x-1)(x-1)^2 + B(1-2x)(x-1) + 2(1-2x)$$

$$7x^2 - 7x + 2 = -2(x-1)(x^2 - 2x + 1) + B(x-1-2x^2+2x) + 2-4x$$

$$7x^2 - 7x + 2 = -2(x^3 - 2x^2 + x - x^2 + 2x - 1) + 2 - 4x$$

$$7x^2 - 7x + 2 = -2x^3 + 4x^2 - 2x + 2x^2 - 4x + 2 + 2 - 4x + B(-2x^2 + 3x - 1)$$

$$7x^2 - 7x + 2 = -2x^3 + 6x^2 - 10x + 4 + B(-2x^2 + 3x - 1)$$

$$7x^2 - 7x + 2 = -2x^3 + 6x^2 - 10x + 4 + B(-2x^2 + 3x - 1)$$

$$2x^3 - 6x^2 + 10x - 2 + 7x^2 - 7x + 2 = B(-2x^2 + 3x - 1)$$

$$\frac{2x^3 + x^2 + 3x}{-2x^2 + 3x - 1} = B$$

$$-2x^2 + 3x - 1$$

$$7x^2 - 7x + 2 = A(x^2 - 2x + 1) + B(1 - 2x)(x - 1) + C(1 - 2x)$$

$$7x^2 - 7x + 2 = \underbrace{(A+B)}_{A=1}(x^2 - 2x + 1) + B(x - 1 - 2x^2 + 2x) - 2(1 - 2x)$$

$$7x^2 - 7x + 2 = x^2 - 2x + 1 + B(-2x^2 + 3x - 1) - 2 + 4x$$

$$7x^2 - 7x + 2 = x^2 - 2x - 1 + B(-2x^2 + 3x - 1)$$

$$7x^2 - 7x + 2 + x^2 - 6x + 3 = B(-2x^2 + 3x - 1)$$

$$8x^2 - 13x + 5 = B(-2x^2 + 3x - 1)$$

$$-2B = 6$$

$$\boxed{B = -3}$$

$$\begin{array}{r} 6x^2 - 5x + 1 \\ - 3x^2 - 2x + 1 \\ \hline 3x^2 - 3x - 2 \end{array}$$

$$G(x) = \frac{1}{1-2x} - \frac{3}{x-1} - \frac{2}{(x-1)^2}$$

$$G(x) = \frac{1}{1-2x} + \frac{3}{1-x} - \frac{2}{(1-x)^2}$$

$$a_r = 2^r + 3 - 2(r+1)$$

$$\boxed{a_r = 2^r - 2^r + 1}$$

6) $a_n - 5a_{n-1} + 6a_{n-2} = 3^n$, $a_0 = 0$, $a_1 = 2$

Solve this recurrence relation by generating function

Ans - Multiply by x^n & sum 2 to ∞

$$\sum_{n=2}^{\infty} a_n \cdot x^n - 5 \sum_{n=2}^{\infty} a_{n-1} \cdot x^n + 6 \sum_{n=2}^{\infty} a_{n-2} \cdot x^n = \sum_{n=2}^{\infty} 3^n \cdot x^n$$

$$(a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots) - 5(a_1 x^2 + a_2 x^3 + a_3 x^4 + \dots) + 6(a_0 x^2 + a_1 x^3 + a_2 x^4 + \dots) = \frac{3^2 x^2}{1-3x}$$

$$G(x) = a_0 + a_1 x + 5x(a_1 x + a_2 x^2 + a_3 x^3 + \dots) + 6x^2(a_0 + a_1 x + a_2 x^2 + \dots) = \frac{1}{1-3x}$$

$$G(x) = 0 + 2x + 5x \cdot [G(x)] + 6x^2 [G(x)] = \frac{1}{1-3x}$$

$$G(x)(6x^2 - 5x + 1) - 2x = \frac{1}{1-3x}$$

$$(6x^2 - 5x + 1)G(x) = \frac{1}{1-3x} + 2x$$

$$(6x^2 - 5x + 1)G(x) = \frac{1 + 2x(1-3x)}{1-3x}$$

$$G(x) = \frac{1 + 2x - 6x^2}{(1-3x)(6x^2 - 5x + 1)}$$

$$G(x) = \frac{-6x^2 + 2x + 1}{(1-3x)(3x-1)(2x-1)}$$

$$\frac{-6x^2 + 2x + 1}{(1-3x)(3x-1)(2x-1)} = \frac{A}{1-3x} + \frac{B}{3x-1} + \frac{C}{2x-1}$$

$$\xrightarrow{R.H.S} \sum_{n=2}^{\infty} 3^n \cdot x^n \quad k(=n-2)$$

$$= \sum_{k=0}^{\infty} 3^{k+2} \cdot x^{k+2}$$

$$= \sum_{k=0}^{\infty} 3^2 \cdot x^2 \cdot \sum_{k=0}^{\infty} 3^k \cdot x^k$$

$$= \frac{9x^2}{1-3x}$$

Unit - 2

PROBABILITY

• Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Ques) A pair of five dice is tossed find the probability of appearance of 3 on one die when the sum is 7?

Ans -

A: To appear 3 on 1 die

B: Sum is 7 (it occurred)

(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)

→ 2 event dependent

$$P(A|B) = \frac{2}{6} = \frac{1}{3}$$

Independent Events

Two events are said to be independent when if the ^{prop} of each is unaffected whether the other happen or not

$$P(A \cap B) = P(A) \times P(B)$$

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

Ques- A box contains 20 electronic components of which 5 are defective if 2 components are selected at random in succession then find the probability that both are defective. When

- When the 1st component is not replaced before the second draw (dependent condⁿ)
- When the 1st component is replaced. (Not dependent condⁿ)

Ans- A: 1st component is selected
B: 2nd component is selected

$$(i) P(A \cap B) = P(A) \times P\left(\frac{B}{A}\right)$$

$$= \frac{5}{20} \times \frac{4}{19}$$

$$= \frac{1}{19}$$

$$(ii) P(A \cap B) = P(A) \times P(B)$$

$$= \frac{5}{20} \times \frac{5}{20} = \frac{1}{16}$$

Ques- 'A' can hit a target 3 times in 5 shots, 'B' two times in five shots & 'C' 3 times in 4 shots. All of them fire one shot each simultaneously at the target. What is the probability that-

(i) 2 shots hit?

E_1 : 'A' hit a shot

E_2 : 'B' hit a shot

E_3 : 'C' hit a shot.

$$P(E_1) = \frac{3}{5}, \quad P(E_2) = \frac{2}{5}, \quad P(E_3) = \frac{3}{4}$$

$$P(2 \text{ shots hit}) = P(E_1, E_2, \bar{E}_3) + P(E_1, \bar{E}_2, E_3) + P(\bar{E}_1, E_2, E_3)$$

$$= P(E_1) \times P(E_2) \times P(\bar{E}_3) + P(E_1) \times P(\bar{E}_2) \times P(E_3) + P(\bar{E}_1) \times P(E_2) \times P(E_3)$$

$$= \frac{3}{5} \times \frac{2}{5} \times \left(1 - \frac{3}{4}\right) + \frac{3}{5} \times \left(1 - \frac{2}{5}\right) \times \frac{3}{4} + \left(1 - \frac{3}{5}\right) \times \frac{2}{5} \times \frac{3}{4}$$

$$= \frac{3}{5} \times \frac{2}{5} \times \frac{1}{4} + \frac{3}{5} \times \frac{3}{5} \times \frac{3}{4} + \frac{2}{5} \times \frac{2}{5} \times \frac{3}{4}$$

$$= \frac{1}{100} (6 + 27 + 12) = \frac{45}{100} = \frac{9}{20}$$

ii) At least 2 shots?

$$= P(2 \text{ shots hit}) + P(E_1) P(E_2) P(E_3)$$

$$= \frac{9}{20} + \frac{3}{5} \times \frac{2}{5} \times \frac{3}{4}$$

$$= \frac{9}{20} + \frac{1}{20} \left(\frac{18}{5} \right)$$

$$= \frac{1}{20} \left(\frac{45+18}{5} \right)$$

45
18
63

$$P(\text{at least 2 shots hit}) = \frac{63}{100}$$

Bayes' Theorem -

If $B_1, B_2, \dots, B_n$ are a set of mutually exclusive events of a sample space S and A is any event associated with $B_1, B_2, \dots, B_n$ such that -

$$A \subseteq \bigcup_{i=1}^n B_i$$

$$P(B_i|A) = \frac{P(B_i) P(A|B_i)}{\sum_{i=1}^n P(B_i) P(A|B_i)}$$

Ques - The contents of bags 1, 2 and 3 are Ist bag contains 1 white 2 black and 3 red balls, II bag contains 2 white, 1 black and 1 red ball, III bag contains 4 white, 5 black and 3 red balls.

One bag is chosen at random and 2 balls are drawn they happen to be white & red. What is the probability that they come from Ist bag?

$$P(B_1) = \frac{1}{3}, P(B_2) = \frac{1}{3}, P(B_3) = \frac{1}{3}$$

A: A event is to choose the 2 balls white & red.

$$P(A|B_1) = \frac{{}^1C_1 \times {}^3C_1}{{}^6C_2} = \frac{1 \times 3}{15} = \frac{1}{5}$$

$${}^n C_r = \frac{n!}{r!(n-r)!}$$

$$P(A/B_2) = \frac{{}^2 C_1 \times {}^1 C_1}{{}^4 C_2} = \frac{1}{3}$$

$$P(A/B_3) = \frac{{}^4 C_1 \times {}^3 C_1}{{}^{12} C_2} = \frac{2}{11}$$

$$P(B_1/A) = \frac{P(B_1) \cdot P(A/B_1)}{P(B_1) \cdot P(A/B_1) + P(B_2) \cdot P(A/B_2) + P(B_3) \cdot P(A/B_3)}$$

$$P(B_1/A) = \frac{\frac{1}{3} \times \frac{1}{5}}{\frac{1}{5} \times \frac{1}{5} + \frac{1}{3} \times \frac{1}{3} + \frac{1}{3} \times \frac{2}{11}}$$

$$= \frac{\frac{1}{5}}{\frac{1}{5} + \frac{1}{3} + \frac{2}{11}} = \frac{\frac{1}{5}}{\frac{33+55+30}{165}} = \frac{\frac{1}{5}}{\frac{118}{165}}$$

$$P(B_1/A) = \frac{\frac{33}{8 \times 118}}{\frac{33}{118}} = \frac{33}{118} \text{ Ans}$$

Ques-³³ A factory produces its entire output with 3 machines. Machine 1, 2 & 3 produces 50%, 30% and 20% of the output. But 4%, 2% & 4% of their outputs are defective respectively. What fraction of the total output is defective.

Ans: $P(B_1) = \frac{50}{100}$, $P(B_2) = \frac{30}{100}$, $P(B_3) = \frac{20}{100}$

↪ A is the defective item.

$$P(A/B_1) = \frac{4}{100}$$

$$P(A/B_2) = \frac{2}{100}, P(A/B_3) = \frac{4}{100}$$

$$P(A) = \sum_{i=1}^n P(B_i) P(A/B_i) = ?$$

$$P(A) = P(A)P(A/B_1) + P(A_2) + P(A/B_2) + P(B_3)P(A/B_3)$$

$$= \frac{50}{100} \times \frac{4}{100} + \frac{30}{100} \times \frac{2}{100} + \frac{20}{100} \times \frac{4}{100}$$

$$= \frac{(20 + 6 + 8)}{100}$$

$$= 20$$

$$= \frac{50}{100} \times \frac{4}{100} + \frac{30}{100} \times \frac{2}{100} + \frac{20}{100} \times \frac{4}{100}$$

$$= \frac{200 + 60 + 80}{100 \times 100}$$

$$= \frac{340}{100 \times 100} = \frac{34}{1000}$$

$$\begin{array}{r} 20 \\ 60 \\ 80 \\ \hline 34 \end{array}$$

$$P(A) = 0.034$$

Random Variables -

• Toss coin 2 times -

$$S = \{HH, HT, TH, TT\}$$

$X \rightarrow$ head (head = 1, probability)

Total sample space = 4

Random variable	X	2	1	0
				(head aye)
P(X)	$\frac{1}{4}$	$\frac{2}{4} = \frac{1}{2}$	$\frac{1}{4}$	

Probability Distribution

There are 2 types of Random Variable -

- 1) Discrete Random Variable
- 2) Continuous Random Variable

Random Variable (RV)

Discrete R.V
 $p(x)$

Continuous R.V
 $f(x)$

probability mass funcⁿ

(i) $p(x_i) \geq 0$

(ii) $\sum p(x_i) = 1$

Mean

$E(x) = \sum x_i p(x_i)$

- 1) Binomial Distribution
- 2) Poisson Distribution

probability density funcⁿ

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

Mean $E(x) = \int_{-\infty}^{\infty} x f(x) dx$

- 1) Exponential Distribution
- 2) Normal Distribution
- 3) Uniform Distribution

Discrete Random Variable

Discrete Distribution Funcⁿ or Cumulative mass Function

The probability that the value of random variable ' X ' is less than or equal to some no. ' x ' is defined as cumulative mass funcⁿ or Distribution Funcⁿ

$$F(x) = P(X \leq x) = \sum p(x_i)$$

Q) A random variable X has the following probability function values -

x	0	1	2	3	4	5	6	7
$P(X=x)$	0	b	$2b$	$2b$	$3b^2$	b^2	$2b^2$	$7b^2+b$

- i) Find the value of b
- ii) Determine the distribution funcⁿ of random variable X
- iii) Evaluate $P(X < 6)$, $P(X \geq 6)$, $P(0 < X < 5)$

Ans (i) - $\sum p(x_i) = 1$

$$0 + b + 2b + 2b + 3b + b^2 + 2b^2 + 7b^2 + b = 1$$

$$9b + 10b^2 - 1 = 0$$

$$10b^2 + 9b - 1 = 0$$

$$10b - 1b$$

$$10b^2 + 10b - b - 1 = 0$$

$$10b^2(b+1) - 1(b+1) = 0$$

$$(10b-1)(b+1) = 0$$

$$10b-1=0$$

$$\left(b = -1 \right), \left(b = \frac{1}{10} \right) \quad \text{Ans (i) } \left(b = \frac{1}{10} \right)$$

we'll neglect it because we know that probability mass function is always greater than 0 i.e. $P(x_i) \geq 0$

$$\text{ii) } P(X=0) = 0$$

$$P(X \leq 1) = P(0) + P(1) \\ = 0 + \frac{1}{10} = \frac{1}{10}$$

$$P(X \leq 2) = P(0) + P(1) + P(2) \\ = 0 + \frac{1}{10} + \frac{2}{10} = \frac{3}{10}$$

$$P(X \leq 3) = P(0) + P(1) + P(2) + P(3) \\ = 0 + \frac{1}{10} + \frac{2}{10} + \frac{2}{10} = \frac{5}{10}$$

$$P(X \leq 4) = P(0) + P(1) + P(2) + P(3) \\ = 0 + \frac{1}{10} + \frac{2}{10} + \frac{2}{10} + \frac{3}{10} = \frac{8}{10}$$

$$P(X \leq 5) = P(0) + P(3) + P(4) \\ = \frac{8}{10} + \frac{1}{100} = \frac{81}{100}$$

$$P(X \leq 6) = P(0) + P(4) + P(5) \\ = \frac{81}{100} + \frac{2}{100} = \frac{83}{100}$$

$$P(X \leq 7) = \frac{83}{100} + \frac{14}{100} + \frac{1}{100} = \frac{100}{100} = 1$$

$$ii) \cdot P(X < 6)$$

$$= P(0) + P(1) + P(2) + P(3) + P(4) + P(5)$$

$$= \frac{81}{100}$$

$$\cdot P(X \geq 6) = P(X=6) + P(X=7)$$

$$= \frac{19}{100}$$

$$\cdot P(0 < X < 5) = P(X=1) + P(2) + P(3) + P(4)$$

$$= \frac{8}{10}$$

Ques- If 'X' is random variable for probability mass funcⁿ as $P(X) = \begin{cases} \frac{x}{15}, & x=1, 2, 3, \dots \\ 0, & \text{otherwise} \end{cases}$

Find

$$i) P(X=1 \text{ or } 2)$$

$$ii) P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right) \quad \text{Conditional probability i.e. this is in form of } P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$\text{Ans(i)} \cdot P(X=1) + P(X=2)$$

$$= \frac{1}{15} + \frac{2}{15} = \frac{3}{15} = \frac{1}{5}$$

$$ii) P\left(\frac{1}{2} < X < \frac{5}{2} \mid X > 1\right) = \frac{P\left(\frac{1}{2} < X < \frac{5}{2} \cap X > 1\right)}{P(X > 1)}$$

$$= \frac{P((X=1, 2) \cap X > 1)}{1 - P(X \leq 1)}$$

$$= \frac{P(X=2)}{1 - P(X=1)} = \frac{\frac{2}{15}}{1 - \frac{1}{15}} = \frac{\frac{2}{15}}{\frac{14}{15}} = \frac{2}{14} = \frac{1}{7}$$

Ans

Continuous Random Variable

Continuous Distribution Form

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx$$

Ques- The diameter of a water pipe, say 'X' is assumed to be a continuous random variable with pdf $f(x)$ = probability density function (P.D.F) -

$$f(x) = 6x(1-x), \quad 0 \leq x \leq 1$$

(i) Check validity of $f(x)$ to be a probability density function

(ii) Determine the value of b , such that $P(X < b) = P(X > b)$

Ans (i) $\int_{-\infty}^{\infty} f(x) dx = 1$ (if we'll get 1 then, it will be a valid P.D.F)

$$\int_{-\infty}^{\infty} 6x(1-x) dx \quad \text{by parts}$$

$$6 \int_{-\infty}^{\infty} (x - x^2) dx$$

$$6 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]$$

$$\int u v dx = u \int v dx - \int u' \left(\int v dx \right) dx$$

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_0^1 6x(1-x) dx \\ &= 6 \int_0^1 [x - x^2] dx \end{aligned}$$

$$= \int_0^1 \left[\frac{x^2}{2} - \frac{x^3}{3} \right] dx$$

$$= 6 \left[\frac{1}{2} - \frac{1}{3} \right] = 6 \left(\frac{3-2}{6} \right)$$

$$= 1$$

∴ it is a probability density function

$$(i) P(X < b) = P(X > b) \longrightarrow (x)$$

$$\int_0^b f(x) dx = \int_b^1 f(x) dx$$

$$\int_0^b (6x - 6x^2) dx = \int_b^1 (6x - 6x^2) dx$$

$$\left[\frac{6x^2}{2} - \frac{6x^3}{3} \right]_0^b = \left[\frac{6x^2}{2} - \frac{6x^3}{3} \right]_b^1$$

$$3b^2 - \frac{2b^3}{3} = 3 - \frac{2}{3} - \left(3b^2 - \frac{2b^3}{3} \right)$$

$$3b^2 + 3b^2 - \frac{2b^3}{3} - \frac{2b^3}{3} = \frac{8}{3}$$

$$6b^2 - \frac{2b^3}{3} = \frac{8}{3}$$

$$18b^2 - 2b^3 = 8$$

$$\div \text{by } 2 \text{ \& } \times \text{ by } -1$$

$$9b^2 - b^3 = -4$$

$$b^3 - 9b^2 + 4 = 0$$

$$b=1$$

$$48-18$$

$$6-9+4$$

$$-1$$

$$-6-9+4$$

$$24$$

$$64$$

$$b^3 - 9b^2 + 4 = 0$$

$$b-2 = \frac{8}{8}$$

$$b = 2$$

$$6b^2 - 9b^3 = \frac{8}{3} \mid \times 3$$

$$18b^2 - 27b^3 = 8$$

$$12b^3 - 18b^2 + 8 = 0$$

$$\div \text{by } 2$$

$$6b^3 - 9b^2 + 4 = 0$$

$$(ii) f(x < a) = f(x > b)$$

$$\int_a^b f(x) dx = \int_b^a f(x) dx$$

$$\int_a^b (6x - 6x^2) dx = \int_b^a (6x - 6x^2) dx$$

$$\left[\frac{6x^2}{2} - \frac{6x^3}{3} \right]_a^b = \left[\frac{6x^2}{2} - \frac{6x^3}{3} \right]_b^a$$

$$3b^2 - 2b^3 = 3 - 2 - 3b^2 + 2b^3$$

$$6b^2 - 4b^3 = 1$$

By Hit & Trial Method

$$6b^2 - 4b^3 = 1 \Rightarrow$$

$$6 \times \frac{1}{2} - 4 \times \frac{1}{8} = 1$$

$$\frac{3}{2} - \frac{1}{2} = \frac{2}{2} = 1$$

$$(b = \frac{1}{2}) \quad 4b^3 - 6b^2 + 1 = 0 \quad 4b^3 - 4b - 2$$

$$\begin{array}{r} 4b^3 - 2b^2 \\ + \\ -4b^2 + 1 \\ -4b^2 + 2b \\ + \\ -2b + 1 \\ -2b + 1 \\ + \\ 0 \end{array}$$

$$a=4, b=1, c=-2$$

$$4b^3 - 4b - 2$$

$$b = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{4 \pm \sqrt{16 + 32}}{2 \times 4 = 8}$$

1) Mean $E(X) = \int_{-\infty}^{\infty} x f(x) dx$ [in term of expectation]
(Continuous variable)

Variance $V(X) = E(X^2) - [E(X)]^2$
 $E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$

Q) Let x be a random variable with pdf is given by - $f(x) = \begin{cases} cx^2, & |x| \leq 1 \\ 0, & \text{otherwise} \end{cases}$

- (i) Find the constant c
 (ii) Find $E(X)$ & $V(X)$
 (iii) Find the probability $x \geq \frac{1}{2}$ i.e. $P(X \geq \frac{1}{2})$

Ans (i) $\int_{-\infty}^{\infty} f(x) dx = 1$ (by pdf property)

$\int_{-1}^1 cx^2 dx = 1$ (as $|x| \leq 1$ range is from -1 to 1 as it is modulus)
 ↓
 even function

$2c \int_0^1 x^2 dx = 1$

$2c \left[\frac{x^3}{3} \right]_0^1 = 1$

$\frac{2c}{3} = 1$

$c = \frac{3}{2}$

i) $E(X) = \int_{-1}^1 \frac{3}{2} x^3 dx = 0$

and $V(x) = E(x^2) - (E(x))^2$ as mean is zero

$$= E(x^2)$$

$$= \frac{3}{2} \int_{-1}^1 x^4 dx = 3 \int_0^1 x^4 dx = \frac{3}{5}$$

$$= \frac{3}{2} \left[\frac{x^5}{5} \right]_{-1}^1$$

$$\int_{1/2}^1 f(x) dx = \frac{3}{2} \int_{1/2}^1 x^2 dx$$

$$= \frac{3}{2} \left[\frac{x^3}{3} \right]_{1/2}^1$$

$$= \frac{3}{2} \left[\frac{1}{3} - \frac{1}{24} \right]$$

$$1 = \frac{2 \cdot 8}{8}$$

$$\frac{8}{8} = 1$$

$$0 = x^3 \cdot \frac{8}{8} \left\{ = f(x) \right\}$$

Q) let X be a continuous random variable with pdf

$$f(x) = \begin{cases} 4x^3, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find $P\left(\frac{x \leq \frac{2}{3}}{x > \frac{1}{3}}\right)$

Ans-
$$\frac{P\left(x \leq \frac{2}{3}\right) \cap P\left(x > \frac{1}{3}\right)}{P\left(x > \frac{1}{3}\right)}$$

$$= \frac{P\left(x \leq \frac{2}{3} \cap x > \frac{1}{3}\right)}{P\left(x > \frac{1}{3}\right)} = \frac{P\left(\frac{1}{3} < x \leq \frac{2}{3}\right)}{P\left(x > \frac{1}{3}\right)}$$

$$= \frac{\int_{1/3}^{2/3} 4x^3 dx}{\int_{1/3}^1 4x^3 dx}$$

$$= \frac{\left[\frac{4x^4}{4}\right]_{1/3}^{2/3}}{\left[\frac{4x^4}{4}\right]_{1/3}^1}$$

$$= \frac{\left[x^4\right]_{1/3}^{2/3}}{\left[x^4\right]_{1/3}^1}$$

$$= \frac{\left(\frac{2}{3}\right)^4 - \left(\frac{1}{3}\right)^4}{1 - \left(\frac{1}{3}\right)^4}$$

$$= \frac{16 - 1}{81 - 1}$$

$$= \frac{15}{80}$$

$$= \frac{3}{16}$$

$$\begin{array}{r} 16 \\ 5 \overline{) 80} \\ \underline{50} \\ 30 \end{array}$$

Ques. Two coins are tossed, what is the expectation of a no. of heads

a coin is tossed -

$$P(X) = \frac{1}{2} \text{ or } P(X)$$

$$E(X) = \sum x f(x)$$

Two heads

$$X = 0, 1, 2, 3, 4$$

$$P(X) = P(0) = \frac{1}{16}$$

$$P(X=1) = \frac{4}{16} = \frac{1}{4}$$

$$P(X=2) = \frac{4 \cdot C_2}{16} = \frac{6}{16}$$

$$P(X=3) = \frac{4 \cdot C_3}{16} = \frac{4}{16}$$

$$P(X=4) = \frac{1}{16}$$

$$E(X) = 0 \times \frac{1}{16} + 1 \times \frac{4}{16} + 2 \times \frac{6}{16} + 3 \times \frac{4}{16} + 4 \times \frac{1}{16}$$

$$= \frac{2 \cdot C_1}{16} \left[\frac{P(X)}{P} \right]$$

Ques. The Random variable Y has probability density function

$f(y)$ is given by -

$$f(y) = \begin{cases} k y (a-y), & 0 \leq y \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

i) Explain why $a \geq 3$

Show that $k = \frac{2}{9(a-2)}$

$$\frac{EY}{EY} = \frac{EY}{EY}$$

$$\frac{1}{1} = \frac{1}{1}$$

Ans (i) -

$$f(y) \geq 0$$

$$ky(a-y) \geq 0$$

$$a-y \geq 0$$

$$413120$$

$$11 \frac{y-1}{2}$$

$$k = 51a - 3000$$

$$a-2 \geq 0$$

$$a \geq 2$$

ii) find the value of k

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_0^a ky(a-y) dy = 1$$

$$k \int_0^a (ay - y^2) dy = 1$$

$$k \left[a \frac{y^2}{2} - \frac{y^3}{3} \right]_0^a = 1$$

$$k \left(\frac{ga}{2} - \frac{27}{3} \right) = 1$$

$$k \left(\frac{ga}{2} - 9 \right) = 1$$

$$k \left(\frac{ga-18}{2} \right) = 1$$

$$k = \frac{1 -}{\frac{ga-18}{2}}$$

$$k = \frac{2}{g(a-2)}$$

(hence proved)

103 x 100 = 10300
130

b) Given that $E(y) = 1.75$
Show that $a = 4$ & write down the value of k

Now $E(y) = 1.75$

$$E(y) = \int_0^a y f(y) dy$$

$$1.75 = \int_0^3 y k y (a-y) dy$$

$$= k \int_0^3 a y^2 (a-y) dy$$

$$1.75 = k \int_0^3 (a y^2 - y^3) dy$$

$$= k \left[\frac{a y^3}{3} - \frac{y^4}{4} \right]_0^3$$

$$1.75 = k \left[\frac{a \times 64}{3} - \frac{256}{4} \right]$$

$$1.75 = k \left[\frac{a \times 4}{3} - \frac{256}{4} \right]$$

$$1.75 = k (21.33)$$

$$(k = 0.082)$$

$$(k = \frac{1}{9})$$

(Answer given)

Binomial Distribution

• With replacement ✓ (use)

A random variable 'x' is said to have binomial distribution if the probability mass function of 'x' is given as

$$P(X=x) = \begin{cases} {}^nC_x p^x q^{n-x} & ; x=0,1,2,\dots \\ 0 & ; \text{otherwise} \end{cases}$$

p = probability of success.

q = probability of failure

n = no. of trials

x = whose probability is required.

Two parameters on which depends on Binomial Distribution:

↳ n & p are two parameters of Binomial Distribution

Ques 1- Assuming half the population of a town consumes chocolates and that 100 investigators each take 10 ~~individuals~~ individuals to see whether they are consumers. How many investigators would you expect to report that 3 people or less were consumers.

Ans. 1 inv

$$p = \frac{1}{2}$$

$$n = 10 \quad (10 \text{ individuals})$$

$$P(X \leq 3) = ?$$

$$\begin{aligned} P(X \leq 3) &= P(X=0) + P(X=1) + P(X=2) + P(X=3) \\ &= {}^{10}C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{10} + {}^{10}C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^9 + {}^{10}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^8 + {}^{10}C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^7 \\ &= \frac{1}{2^{10}} \left(1 + 10 + 45 + 120 \right) \\ &= \frac{1}{1024} (1 + 10 + 45 + 120) \end{aligned}$$

$$P(X \leq 3) = \frac{176}{1024}$$

$$P(X \leq 3) = \frac{88}{512}$$

$$\text{Total investigators} = \frac{176}{1024} \times 100$$

$$\text{Mean} = n \times p, \quad \text{Variance} = n \times p \times q \quad \left\{ \begin{array}{l} \text{in Binomial} \\ \text{Distribution} \end{array} \right.$$

Ques 2 - A & B play a game in which their chances of winning are in ratio 3:2. Find A's chance of winning atleast 3 games out of 5 games played.

→ either will win or lose (Binomial distribution)

ratio 3:2

$$n = 5, \quad p = \frac{3}{5} \rightarrow A(\text{win}), \quad q = 1 - \frac{3}{5} = \frac{2}{5}$$

$$P(X \geq 3) = ?$$

$$P(X \geq 3) = P(X=3) + P(X=4) + P(X=5)$$

$$= {}^5C_3 \left(\frac{3}{5} \right)^3 \left(\frac{2}{5} \right)^2 + {}^5C_4 \left(\frac{3}{5} \right)^4 \left(\frac{2}{5} \right)^1 + {}^5C_5 \left(\frac{3}{5} \right)^5$$

$$= 10 \times \left(\frac{3}{5} \right)^3 \left(\frac{2}{5} \right)^2 + 5 \times \left(\frac{3}{5} \right)^4 \left(\frac{2}{5} \right)^1 + \left(\frac{3}{5} \right)^5$$

$$= \frac{10 \times 27 \times 4}{3125} + 5 \times 81 \times 2$$

$$= \frac{1080}{3125} + \frac{810}{3125} + \frac{243}{3125}$$

$$= \frac{2133}{3125} = 0.682$$

Poisson Distribution -

if n in any exp 'n' is very large & p probability is very less

→ A random variable 'x' is said to follow Poisson Distribution if it assumes only non-negative values & its probability mass function is given by -

$$P(X=x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & x=0,1,2,\dots, \lambda>0 \\ 0, & \text{otherwise.} \end{cases}$$

$$\lambda = np$$

• λ is one parameter of Poisson Distribution.

$$\text{Mean} = \lambda, \quad \text{Variance} = \lambda$$

Q1) Suppose that a book of 600 pages contains 40 printing mistakes assume that these errors are randomly distributed throughout the book & 'x' the no. of errors per page has a Poisson distribution. what is the probability that 10 pages selected at random will be free of errors.

Ans - $n = 10, \quad p = \frac{40}{600}, \quad x = \text{no. of errors}$

$$P(X=0) = ?$$

$$P(X=0) = \frac{e^{-0.666}}{0!}$$

$$\lambda = np = 10 \times \frac{40}{600}$$

$$(\lambda = 0.666)$$

$$P(X=0) = \frac{e^{-0.666} (0.666)^0}{0!} \\ = e^{-0.666}$$

Q). Find the probability that almost 5 defective fuses will be found in a box of 200 fuses if experiences show that 2% of such fuses are defective.

$$(e^{-4} = 0.0183)$$

Ans -

$$n = 5$$

$$p = \frac{4}{200} \quad (\text{defective})$$

$$\rightarrow \frac{(200 \times 2\% = 4) = 4}{200}$$

$$\lambda = np$$

$$\lambda = 200 \times \frac{4}{200}$$

$$200 \times \frac{2}{100}$$

$$(\lambda = 4)$$

$$P(X \leq 5) = P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= e^{-4.0000} (\lambda^1 + \lambda^2 + \lambda^3 + \lambda^4 + \lambda^5)$$

$$= e^{-4.0000} \left(\frac{4^1}{0!} + \frac{4^2}{1!} + \frac{4^3}{2!} + \frac{4^4}{3!} + \frac{4^5}{4!} \right)$$

$$= e^{-4.0000} (1 + 16 + 64 + 256 + 1024)$$

$$= e^{-0.0183} \left(\frac{0.04}{0!} + \frac{0.16}{1!} + \frac{0.64}{2!} + \frac{2.56}{3!} + \frac{10.24}{4!} \right)$$

$$= e^{-4.0000} \left(1 + 4 + 8 + \frac{64}{6} + \frac{256}{24} + \frac{1024}{120} \right)$$

$$= e^{-0.0183} (13 + 10.66 + 10.666 + 8.533)$$

$$= e^{-4} (42.853)$$

$$= 0.0183 \times 42.853$$

$$P(X \leq 5) = 0.7842$$

Q2) A coffee sampler claims that he can distinguish b/w a cup of instant coffee & a cup of percolator coffee 75% of the time. It is agreed that his claim will be accepted if he correctly identifies atleast 5 of the 6 cups.

Find his chance of having the claim.

(i) accepted (ii) rejected, when she does have the ability to claim.

Ans 2 -

$$\frac{P = \frac{153}{1504}}{P = \frac{3}{4}} \quad n=6, \quad n=5$$

$$\begin{aligned} i) P(X \geq 5) &= P(X=5) + P(X=6) \\ &= {}^6C_5 \left(\frac{3}{4}\right)^5 \left(\frac{1}{4}\right)^1 + {}^6C_6 \left(\frac{3}{4}\right)^6 \left(\frac{1}{4}\right)^0 \end{aligned}$$

$$= \frac{6!}{8!} = \frac{1}{8}$$

$$P(X \geq 5) = 0.109$$

Rejected:

$$= 1 - P(X \geq 5)$$

$$= 1 - 0.09$$

$$= 0.891$$

Unit - 5

$G(V, E)$ i.e. A graph 'G' consists of a finite set of vertices 'V' and finite set of edges 'E'.

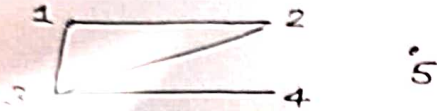
Two types of graph-

i) Directed graph (digraph) ii) Undirected graph

$$E = \{ (v_i, v_j) : v_i, v_j \in V \}$$

$$V = \{ 1, 2, 3, 4, 5 \}$$

$$E = \{ (1, 2), (2, 3), (1, 3), (3, 4) \}$$

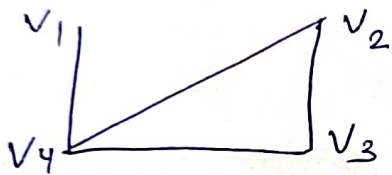


This is a graph
(Undirected graph $\rightarrow$ no direction)

Size of graph = no. of vertices = $|V| = 5$

Size of graph - no. of edges = $|E| = 4$

Adjacent Vertices:



V_1 and V_4

V_2 and V_4

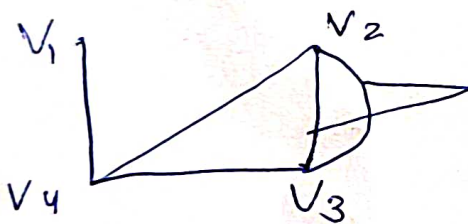
V_4 and V_3

V_2 and V_3

$\rightarrow$ adjacent vertices
(connected to one edge)

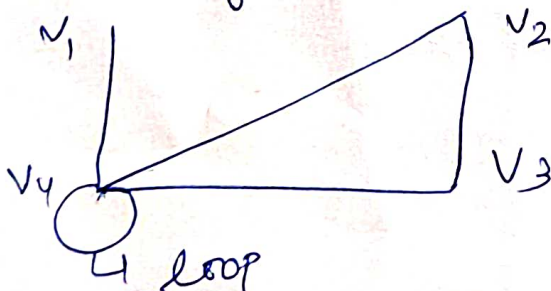
Parallel edges:

Two vertices having more than one edges



Parallel edges

Loop: Starting and ending vertex and same

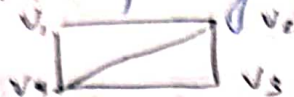


Kinds of Graphs

1) Simple Graph:

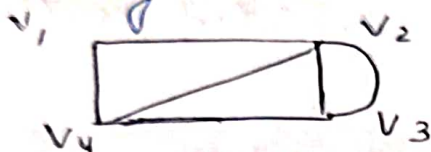
A graph 'G' that has no loop or parallel edges is called a simple graph.

eg



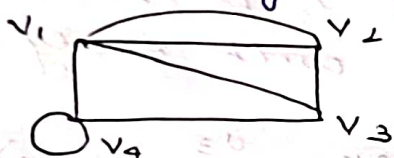
2) Multi Graph-

A graph G is known as the multigraph if it contains parallel edges but not the loops.



3) Pseudo Graph-

A graph G is known as the Pseudograph if we allow both parallel edges and loop is called Pseudo Graph.

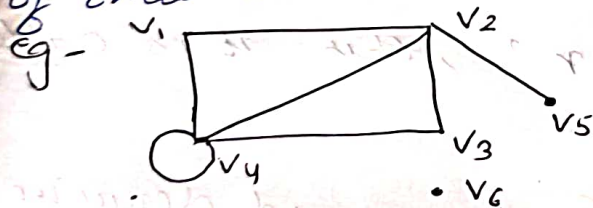


Degree of Vertex:

i) Undirected Graph:

The degree of a vertex of an undirected graph is the no. of edges incident with it except that a loop at a vertex contributes twice to the degree of that vertex.

that vertex degree is zero $\rightarrow$ (isolated vertex)



$$\deg_G(v_1) = 2, \deg_G(v_6) = 0$$

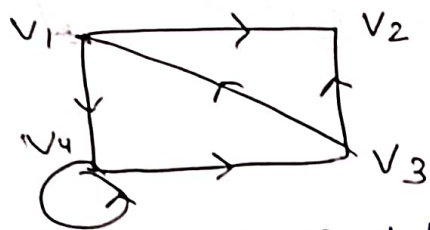
$$\deg_G(v_4) = 5$$

$$\deg_G(v_5) = 1 \text{ (Pendant vertex)}$$

that vertex degree is one.

ii) Directed Graph:

In the directed graph 'G' the outdegree of the vertex is the no. of edges beginning at that vertex & the indegree of the vertex is the no. of edges ending at that vertex. And the sum of these degree is the directed graph.

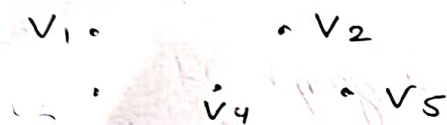


(loop both direction) $\rightarrow$ directed graph.

Types of Graph:

1) Null Graph:-

A graph which contains only isolated vertices is called a Null Graph.



$N_5 = 5$ vertices

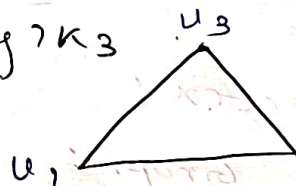
Complete Graph - (K_n)

A simple graph 'G' is said to be complete if every vertex of 'G' is connected with every other vertices.

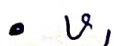
eg) K_4



eg) K_3



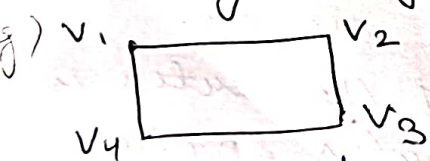
eg) K_1



2) Regular Graph:

A graph in which all vertices are of equal degree is called a regular graph.

If any graph has degree 'r', then it is called r-regular graph.



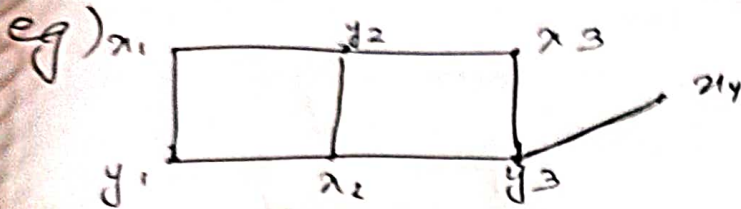
2-regular graph

(as all vertex has equal degree is 2)

Every complete graph is a regular graph but vice-versa is not true.

4) Bipartite Graph-

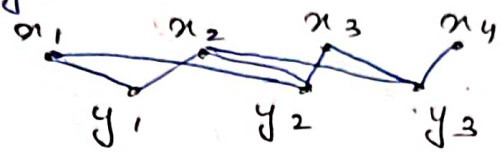
A graph 'G' is Bipartite if the vertex set 'V' can be partitioned into two subsets such that every edge in 'E' connects the vertex in 'V₁' and a vertex in 'V₂'. ~~such~~ that no vertex in 'G' connects either two vertices in 'V₁' and two vertices in 'V₂'.
(V₁, V₂) is called the Bipartition of G



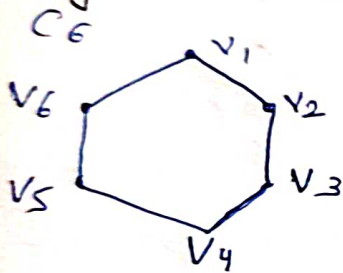
$$V = \{x_1, y_1, x_2, y_2, x_3, y_3, x_4\}$$

$$V_1 = \{x_1, x_2, x_3, x_4\}$$

$$V_2 = \{y_1, y_2, y_3\}$$



5) Cyclic Graph - (where starting & ending both are same)



$$V_1 = \{V_1, V_3, V_5\}$$

$$V_2 = \{V_2, V_4, V_6\}$$

(C_6 is the Bipartite Graph)